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Buffer for holding a pipe

Abstract

A buffer for holding a pipe adapted to transport a fluid includes a base. The buffer further includes a plurality of fingers extending outwardly from a first side of the base, wherein a first finger of the plurality of fingers and a second finger of the plurality of fingers define a cavity for receiving the pipe, the first finger extends outwardly from a first end of the base, and the second finger extends outwardly from a central portion of the base, and a third finger of plurality of fingers extends outwardly from a second end of the base opposite the first end, wherein the second finger is between the first finger and the third finger. The buffer further includes at least one roller on a second side of the base opposite the first side.

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Background/Summary

PRIORITY CLAIM (1) This application is a continuation of U.S. application Ser. No. 17/853,577, filed Jun. 29, 2022, now U.S. Pat. No. 11,754,469, issued Sep. 12, 2023, which is a divisional of U.S. application Ser. No. 16/595,045, filed Oct. 7, 2019, now U.S. Pat. No. 11,635,347, issued Apr. 25, 2023, which claims priority to U.S. Provisional Patent Application Ser. No. 62/753,376 filed Oct. 31, 2018, which are incorporated herein by reference in their entireties.

BACKGROUND

(1) In the manufacture of integrated circuits, hundreds of processing steps, including deposition, photolithography, chemical mechanical planarization (CMP), ion implantation, diffusion, etching

and cleaning, are used to fabricate circuit components on a semiconductor wafer. Numerous semiconductor processing tools are utilized during the fabrication of integrated circuits.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a perspective view of a buffer that is usable to reduce friction-induced wear of a pipe for transporting a fluid, in accordance with some embodiments.
- (3) FIG. 2 is a perspective view of a buffer holding a fluid-transporting pipe and an electrical cable over a support plate, in accordance with some embodiments.
- (4) FIG. 3 is a schematic diagram of a system for monitoring the wear of one or more rollers in the buffer over time, in accordance with some embodiments.
- (5) FIG. 3A is a plot illustrating changes in acoustic wave intensity as a buffer wears, in accordance with some embodiments.
- (6) FIG. 4 is a schematic diagram of an arrangement in a semiconductor fabrication facility using a buffer, in accordance with some embodiments.
- (7) FIG. 5 is a flow chart of a method for monitoring the wear of one or more rollers in a buffer, in accordance with some embodiments.
- (8) FIG. 6 is a schematic diagram of a controller system, in accordance with some embodiments

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, values, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. System may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(11) Pipes such as plastic pipes made of Teflon™ fluoropolymers are widely used to distribute fluids throughout a semiconductor fabrication facility. Fluoropolymers are chemically inert, thus reducing the risk of contamination and on-wafer defects associated with metal pipes. In some instances, in various fabrication steps for formation of integrated circuits, process fluids, including process gasses and process liquids, are utilized. Process fluids are stored in tanks and are transported from tanks to semiconductor processing tools in which the process fluids are used by

way of pipes. In some instances, in fabrication of integrated circuits, various fabrication processes are conducted in semiconductor processing tools. These semiconductor processing tools include processing chambers used in chemical vapor deposition (CVD), physical vapor deposition (PVD), or growing native oxides such as silicon oxide. Components inside the processing chambers are heated up during the deposition operation. To maintain temperature in the processing chambers, a cooling system is provided to control the temperatures of processing chambers. The cooling system circulates a cooling liquid such as water or a mixture of water and glycol through pipes to remove heat from the processing chambers.

(12) Initially, a pipe is maintained in a state of being aligned in a straight line. However, with the increasing time duration in use of the pipe, the pipe ages. Sagging or drooping of the pipe occurs due to gravity or other forces, and excessive sagging or drooping of the pipe causes the pipe to touch a support plate (e.g., a metal plate) where the semiconductor processing tools are placed. As a fluid is transported through the pipe, the pipe vibrates and rubs against the surface of the support plate which has a high coefficient of friction. Over time, the friction between the pipe and the surface of the support plate causes the pipe to break, resulting in a leakage of the fluid that flows through the pipe. The fluid leakage exposes workers to the leaked fluid or causes shorts of electronic components in the semiconductor fabrication facility, thus compromising the fabrication of integrated circuits.

(13) A buffer is provided to help to prevent the friction-induced damages to the fluid-transporting pipes used in a semiconductor fabrication facility. The buffer is configured to support and elevate a fluid-transporting pipe, thereby helping to prevent the fluid-transporting pipe from rubbing against a support plate where semiconductor processing tools are placed. The buffer includes a material having a coefficient of friction lower than the coefficient of friction of the surface material of the support plate so as to provide a low friction contact between the pipe and the buffer. Separating the fluid-transporting pipe from the support plate using a low friction buffer thus helps to improve the durability of the fluid-transporting pipe. One or more rollers are provided at the bottom of the buffer which allow the buffer to move in any direction on the support plate. The one or more rollers reduce the contact area between the buffer and the support plate, thereby helping to reduce the friction between the buffer and the support plate. As a result, the lifetime of the buffer is increased. The wear of the one or more rollers is monitored by an acoustic sensor configured to detect sound waves generated as the one or more rollers move on the support plate. When an abnormal sound wave caused by the wear of the one or more rollers is detected by the acoustic sensor, an operator is notified to determine when to replace the worn rollers before any damage to the pipe occurs.

(14) FIG. 1 is a perspective view of a buffer **100** that is usable to hold a pipe so as to reduce friction-induced cracking of the pipe, in accordance with some embodiments. In some embodiments, the pipe is adapted to transport a fluid in a fluid supply system in a semiconductor fabrication facility.

(15) Referring to FIG. 1, the buffer **100** includes a base **110**, a plurality of fingers (e.g., fingers **122**, **124** and **126**) extending outwardly from one side of the base **110** and spaced from each other, and one or more rollers **152** on an opposite side of the base **110** from the plurality of fingers **122**, **124** and **126**. The number of the rollers **152** is determined by the size of the base **110**. In some embodiments, the plurality of fingers **122**, **124** and **126** and one or more rollers **152** have a unitary construction with the base **110**, i.e., the base **110**, the plurality of fingers **122**, **124** and **126**, and the one or more rollers **152** being integral with each other.

(16) In some embodiments, the plurality of fingers includes a first finger **122** extending outwardly from a first end **110a** of the base **110**, a second finger **124** extending outwardly from a central portion **110b** of the base **110**, and a third finger **126** extending outwardly from a second end **110c** of the base **110** opposite the first end **110a**. The first finger **122** is configured to have a length less than a length of each of the second finger **124** and the third finger **126**. In some embodiments, the second finger **124** and the third finger **126** have a same length. In some embodiments, the second

finger **124** and the third finger **126** have different lengths. The first finger **122** includes a ridge **132** protruding from a free end **122a** of the first finger **122** and toward the second finger **124**. The second finger **124** includes a ridge **134** protruding from the second finger **124** and toward the first finger **122**. The ridges **132** and **134** face each other and define an open end **136a** of a cavity **136** formed by the first finger **122**, a first segment of the base **110** (herein referred to as a first base segment **112**), and a portion of the second finger **124**. The cavity **136** is configured to receive a pipe **142** (FIG. 2) for transporting a fluid (also referred to as fluid-transporting pipe) when the buffer **100** is in use. In some embodiments, the cavity **136** is C-shaped having a curved inner surface. The dimension of the cavity **136** is determined based on the size of the fluid-transporting pipe **142** to be held therein. In some embodiments, the cavity **136** is dimensioned to fit a one-inch pipe. In some embodiments, the cavity **136** is dimensioned to fit a three-quarter-inch pipe. In some embodiments, the cavity **136** has greater or lesser dimensions for different diameter pipes. The open end **136a** of the cavity **136** from which the fluid-transporting pipe **142** is inserted is configured to have a dimension less than a diameter of the fluid-transporting pipe **142** such that once the fluid-transporting pipe **142** is forced into the cavity **136**, the cavity **136** holds the fluid-transporting pipe **142** firmly in place. The dimension of the open end **136a** of the cavity **136** is determined by the heights of the ridges **132** and **134**. The second finger **124**, the third finger **126**, and a second segment of the base **110** (herein referred to as second base segment **114**) define a cavity **138**. In some embodiments, the second base segment **114** has a curved surface. The dimension of the cavity **138** is configured to accommodate an electrical cable **144** (FIG. 2) when the buffer **100** is in use. In some embodiments, the cavity **138** has a uniform dimension that is less than a diameter of the electrical cable **144** such that once the electrical cable **144** is forced into the cavity **138**, the electrical cable **144** is pressed by the second finger **124** and the third finger **126**, which prevents the electrical cable **144** from sliding out of the cavity **138**. In some embodiments, the cavity **138** is slanted such that the dimension of the cavity **138** reduces gradually towards the free ends of the second finger **124** and the third finger **126**. In some embodiments, an open end **138a** of the cavity **138** is configured to have a dimension less than the diameter of the electrical cable **144**. The smaller dimension of the open end **138a** of the cavity **138** helps to prevent the electrical cable **144** from sliding out of the cavity **138** once the electrical cable **144** is received into inner portion of the cavity **138**. The third finger **126** is optional, and in some embodiments, the third finger **126** is omitted. In the embodiments where the third finger **126** is omitted, the second finger **124** is present at the second end **110c** of the base **110**.

(17) The one or more rollers **152** are protrusions protruding from the base **110** of the buffer **100** and are adapted to allow the buffer **100** to move in any direction. The one or more rollers **152** are configured to reduce the friction between the buffer **100** and a support plate **102** (FIG. 2) over which the buffer **100** is placed. In some embodiments, each roller **152** of the one or more rollers **152** is a hemispherical-shaped protrusion having a rounded bottom side adapted to contact the support plate **102** when the buffer **100** is in use. The rounded bottom side reduces the contact area between the buffer **100** and the support plate **102**, thereby helping to reduce the friction between the buffer **100** and the support plate **102** and to increase the durability of the buffer **100**. The buffer **100** also includes a ring **154** surrounding each roller **152**. The ring **154** is optional, and in some embodiments, the ring **154** is omitted.

(18) In some embodiment, the buffer **100** includes a material having a lower coefficient of friction than the material of the underlying support plate. As used herein, coefficient of friction refers to a kinetic coefficient of friction which is defined as the ratio of the normal force required to maintain a steady state motion of an object sliding on a given surface. In some embodiments, the support plate includes metal such as iron. In some embodiments, the support plate includes wood or plywood. In some embodiments, the coefficient of friction of the low friction material of the buffer **100** is less than 0.1. In some embodiments, the buffer **100** includes polytetrafluoroethylene (PTFE), perfluoroalkoxy (PFA), fluorinated polyethylene propylene (FPEP), polyvinylidene fluoride

(PVDF), polysulfone, or polyether ether ketone (PEEK).

(19) FIG. 2 is a perspective view of a buffer **100** holding a fluid-transporting pipe **142** and an electrical cable **144** over a support plate **102**, in accordance with some embodiments. In FIG. 2, the buffer **100** is configured to hold the fluid-transporting pipe **142** and the electrical cable **144** in the cavities **136** and **138**, respectively. The buffer **100** thus acts as a cushion to avoid the direct contact of the fluid-transporting pipe **142** and the electrical cable **144** with the support plate **102**, which helps to eliminate the effect of the high friction force applied on the fluid-transporting pipe **142** and the electrical cable **144** by the support plate **102**. The low friction material used in the buffer **100** provides low friction contact surfaces when the fluid-transporting pipe **142** and the electrical cable **144** are received in the cavity **136** and the cavity **138** of the buffer **100**, respectively. As a result, the friction-induced wear of the fluid-transporting pipe **142** or the electrical cable **144** is reduced, and the usable time of the fluid-transporting pipe **142** or the electrical cable **144** is increased.

(20) FIG. 3 is a schematic diagram of a system **300** for monitoring the wear of one or more rollers **152** in the buffer **100** over time, in accordance with some embodiments.

(21) Referring to FIG. 3, the system **300** includes an acoustic sensor **310** and a sensor control unit **320**. The acoustic sensor **310** is adapted to detect acoustic waves generated during the use of the buffer **100** due to the vibration of the one or more rollers **152** caused by the friction force between the one or more rollers **152** and the support plate **102**. As a roller **152** wears, the height of the roller **152** reduces due to the friction between the roller **152** and the support plate **102** (FIG. 2).

Consequently, the contact area between the roller **152** and the support plate **102** increases with time. The frequencies of the acoustic waves measured by the acoustic sensor **310** thus change over time with the increased contact area between the roller **152** and the support plate **102**. The changes in frequencies of the acoustic waves are usable to evaluate the amount of the wear of the roller **152**.

(22) FIG. 3A is a plot illustrating changes in acoustic wave intensity as a buffer **100** wears, in accordance with some embodiments. Curve **340** illustrates an initial acoustic wave intensity of a buffer **100** as a function of time. Curve **350** illustrates an acoustic wave intensity of the buffer **100** as a function of time after the roller **152** starts wearing when the buffer **100** has been used for a certain period of time. Compared to the curve **340** of the intensity of the initial acoustic wave, after the buffer **100** has been used for a certain period of time, the curve **350** of the intensity of the acoustic wave increases about 10 hertz (Hz).

(23) The acoustic sensor **310** is placed close to the buffer **100** such that acoustic waves originated from the abrasive force between the one or more rollers **152** and the support plate **102** are able to be received by the acoustic sensor **310**. In some embodiments, the acoustic sensor **310** is placed about 30 centimeters (cm) away from the buffer **100**. If the distance between the acoustic sensor **310** and the buffer **100** is too great, the risk that the acoustic sensor **310** is not able to detect the acoustic waves increases. In some embodiments, the acoustic sensor **310** is an ultrasonic sensor usable to detect sound waves that are beyond an audible range of frequency to human ears, typically above 20 kilohertz (kHz). The ultrasonic sensor operates at high frequencies, and thus has high sensitivity in detecting the acoustic waves generated by the contact between the one or more rollers **152** and the support plate.

(24) The sensor control unit **320** is adapted to analyze changes of frequencies of the acoustic waves detected by the acoustic sensor **310**. Once an intensity of acoustic wave signals exceeds a threshold value that is associated with a normal operating condition of the one or more rollers **152**, an operator is notified to replace the buffer **100** to avoid the excess abrasion between the buffer **100** and the support plate **102**. In some embodiments, the threshold value is set to be 3300 Hz. In some embodiments, the sensor control unit **320** is a multichannel control unit. The sensor control unit **320** communicates wirelessly, for instance, via a communication network **330**. In some embodiments, the sensor control unit **320** communicates via a wired connection with acoustic sensor **310**. In some embodiments, the sensor control unit **320** is implemented by a controller system **600** (FIG. 6), and the communication network **330** is implemented by a network interface

608 in the controller system **600** (FIG. 6).

(25) FIG. 4 is a schematic diagram of an arrangement **400** in a semiconductor fabrication facility using a buffer **100** of FIG. 1 to reduce the friction-induced wear of a pipe for transporting a fluid, in accordance with some embodiments.

(26) Referring to FIG. 4, the arrangement **400** includes a semiconductor processing system **410**, a fluid supply system **420**, a power supply **430**, and a controller system **600**.

(27) The semiconductor processing system **410** includes at least one processing chamber **412**. The at least one processing chamber **412** is designed to perform one or more semiconductor manufacturing processes applied to one or more semiconductor wafers. In some embodiments, the processing chamber **412** is designed to perform the semiconductor manufacturing process, such as, deposition, thermal oxidation, implantation, lithography exposure, ion implantation, or etching. In some embodiments, the processing chamber **412** is a deposition tool, such as a chemical vapor deposition (CVD) tool or a physical vapor deposition (PVD) tool. In some embodiments, the processing chamber **412** is a CVD tool usable to form a dielectric layer on a semiconductor substrate for isolation. In some embodiments, the processing chamber **412** is a PVD tool usable to form a metal layer for interconnection. In some embodiments, the processing chamber **412** is an ion implantation tool usable to perform an ion implantation process for forming one or more doped features, such as source/drain regions or N-type or P-type wells, in a semiconductor wafer. In some embodiments, the processing chamber **412** is a chemical mechanical polishing (CMP) tool usable to polish a semiconductor wafer to reduce the thickness variation and provide a planarized surface. In some embodiments, the processing chamber **412** is a lithography tool usable to expose a photoresist layer on a semiconductor wafer using a radiation energy in order to form the patterned photoresist layer in assistance of other processing steps, such as etching, deposition, or ion implantation. In some embodiments, the semiconductor processing system **410** is a cluster system having multiple processing chambers configured to perform a same processing function or different processing functions (not shown). The processing chamber **412** shown in FIG. 4 is one processing chamber of the plurality of processing chambers in the cluster tool. In some embodiments, some processing chambers in the plurality of processing chambers are adopted for deposition of different materials, such as, for example, titanium nitride (TiN), titanium (Ti), and aluminum (Al), and some of processing chambers in the plurality of processing chambers are adopted for degassing, pre-cleaning, and cooling.

(28) The fluid supply system **420** is adapted to supply a fluid for use in semiconductor processing operations. In some embodiments, the fluid supply system **420** supplies a slurry to the processing chamber **412** for CMP operation. In some embodiments, the fluid supply system **420** supplies a process gas or a process liquid to the processing chamber **412** for deposition operation. In some embodiments, the fluid supply system **420** supplies water to the processing chamber **412** for cooling components of the processing chamber **412**.

(29) The fluid is supplied to the processing chamber **412** via a fluid-transporting pipe **422**. The fluid-transporting pipe **422** is supported by a buffer **100**. In some embodiments, the fluid-transporting pipe **422** is made of a chemically inert material such as, for example, polytetrafluoroethylene (PTFE). In some embodiments, the fluid-transporting pipe **422** is provided as a linear pipe for transporting a process fluid to the processing chamber **412**. In some embodiments, the fluid-transporting pipe **422** is provided as a coil-type pipe for distributing a cooling liquid to the processing chamber **412** or to a heat source (not shown).

(30) The power supply **430** is usable to provide power to components of the semiconductor processing system **410**, the fluid supply system **420** and the acoustic sensor **310**. The power supply **430** is electrically connected to the semiconductor processing system **410**, the fluid supply system **420**, and the acoustic sensor **310** by respective electrical cables **432**, **434** and **436**. In some embodiments, the electrical cable **432** that electrically connects the semiconductor processing system **410** to the power supply **430** is received in the cavity **138** of the buffer **100** (FIG. 1).

(31) The controller system **600** (described in detail in FIG. 6) is adapted to control the operations of semiconductor processing system **410**, the fluid supply system **420** and the acoustic sensor **310**.

(32) The buffer **100** separates the fluid-transporting pipe **422** and the electrical cable **432** from a support plate over which the fluid-transporting pipe **422** and the electrical cable **432** are placed. The buffer **100** is made of a material having a lower coefficient of friction than the coefficient of friction of the support plate, thereby helping to reduce the friction-induced wear of the fluid-transporting pipe **422** and the electrical cable **432**. Using a buffer **100** increases the usable lifetime of the fluid-transporting pipe **422** and the electrical cable **432**. As a result, downtime of the semiconductor processing system **410** caused by the changing of the worn pipe is reduced. The buffer **100** also separates the electrical cable **432** from the support plate, and thus helps to reduce the risk of shorts caused by the leaked liquid when the fluid-transporting pipe **422** breaks.

(33) FIG. 5 is a flow chart of a method **500** for monitoring wear of one or more rollers **152** in a buffer **100**, in accordance with some embodiments. In some embodiments, additional processes are performed before, during, and/or after the method **500** in FIG. 5, and some of processes described herein are replaced or eliminated in some embodiments.

(34) In operation **502**, a buffer **100** is provided to separate a fluid-transporting pipe **142** and/or an electrical cable **144** from a support plate **102**. The support plate **102** is made of a material having a coefficient of friction higher than the coefficient of friction of the material that provides the buffer **100**. In some embodiments, the fluid-transporting pipe **142** and the electrical cable **144** are connected to a semiconductor processing system **410** in a semiconductor fabrication facility.

(35) In operation **504**, acoustic waves generated by the vibration of one or more rollers **152** provided at the bottom of the buffer **100** are continuously monitored using an acoustic sensor **310**.

(36) In operation **506**, changes in frequencies of the acoustic waves over time are analyzed using a sensor control unit **320** to determine the wearing status of the one or more rollers **152**.

(37) In operation **508**, after an increase of acoustic wave intensity due to the wear of the one or more rollers **152** exceeds a threshold value that is associated with a normal working condition of the one or more rollers **152**, the sensor control unit **320** triggers an alert to notify an operator that the buffer **100** is worn and needs to be replaced. In some embodiments, when the increase of intensity of the acoustic wave is above 3300 Hz, the alert is triggered.

(38) FIG. 6 is a schematic diagram of a controller system **600**, in accordance with some embodiments. The controller system **600** generates output control signals for controlling operations of the processing chamber(s) **412** and other components of semiconductor processing system **410**, the fluid supply system **420**, the power supply **430**, and the acoustic sensor **310**, in accordance with some embodiments. The controller system **600** receives input signals from the processing chamber(s) **412** and other components of semiconductor processing system **410**, the fluid supply system **420**, the power supply **430**, and the acoustic sensor **310**, in accordance with some embodiments.

(39) The controller system **600** includes a processor **602**, an input/output (I/O) device **604**, a memory **606**, and a network interface **608** each communicatively coupled via a bus **610** or other interconnection communication mechanism.

(40) The processor **602** is arranged to execute and/or interpret one or more set of instructions **612** stored in the memory **606**. In some embodiments, the processor **602** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

(41) The I/O device **604** is coupled to external circuitry. In some embodiments, the I/O interface **604** includes a keyboard, keypad, mouse, trackball, trackpad, and/or cursor direction keys for communicating information and commands to the processor **602**.

(42) The memory **606** (also referred to as a computer-readable medium) includes a random access memory or other dynamic storage device, communicatively coupled to the bus **610** for storing data and/or instructions for execution by the processor **602**. In some embodiments, the memory **606** is

used for storing temporary variables or other intermediate information during execution of instructions to be executed by the processor **602**. In some embodiments, the memory **606** also includes a read-only memory or other static storage device coupled to the bus **610** for storing static information and instructions for the processor **602**. In some embodiments, the memory **606** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, the memory **606** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In some embodiments using optical disks, the memory **606** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

(43) The memory **606** is encoded with, i.e., storing, the computer program code, i.e., a set of executable instructions **612**, for controlling one or more components of the semiconductor processing system **410**, the fluid supply system **420**, the power supply **430**, and the acoustic sensor **310** and causing the controller system **600** to perform the method **500**. In some embodiments, the memory **606** also stores information needed for performing the method **500** as well as information generated during performing the method **500**.

(44) The network interface **608** includes a mechanism for connecting to a network **609**, to which one or more other computer systems are connected. In some embodiments, the network interface **608** includes a wired and/or wireless connection mechanism. The network interface **608** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interface such as ETHERNET, USB, or IEEE-1394. In some embodiments, the controller system **600** is coupled with one or more components of the semiconductor processing system **410**, the fluid supply system **420**, the power supply **430**, and the acoustic sensor **310** via the network interface **608**. In some embodiments, the controller system **600** is directly coupled with one or more components of the semiconductor processing system **410**, the fluid supply system **420**, the power supply **430**, and the acoustic sensor **310**, e.g., with the components coupled to the bus **610** instead of via the network interface **608**.

(45) An aspect of this description relates to a buffer for holding a pipe adapted to transport a fluid. The buffer includes a base. The buffer further includes a plurality of fingers extending outwardly from a first side of the base, wherein a first finger of the plurality of fingers and a second finger of the plurality of fingers define a cavity for receiving the pipe, the first finger extends outwardly from a first end of the base, and the second finger extends outwardly from a central portion of the base, and a third finger of plurality of fingers extends outwardly from a second end of the base opposite the first end, wherein the second finger is between the first finger and the third finger. The buffer further includes at least one roller on a second side of the base opposite the first side. In some embodiments, a length of the first finger is less than a length of at least one of the second finger or the third finger. In some embodiments, the first finger includes a first protrusion extending from an end of the first finger distal from the base, wherein the first protrusion extends toward the second finger. In some embodiments, the second finger includes a second protrusion extending from a top surface of the second finger, and the second protrusion extends toward the first finger. In some embodiments, the second protrusion is aligned with the first protrusion. In some embodiments, the second finger and the third finger define a second cavity therebetween. In some embodiments, the second cavity comprises a curved wall adjacent to the base. In some embodiments, the buffer further includes a ring surrounding the at least one roller. In some embodiments, the at least one roller includes a plurality of rollers.

(46) An aspect of this description relates to a buffer for holding a pipe adapted to transport a fluid. The buffer includes a base. The buffer further includes a plurality of fingers extending outwardly from a first side of the base. The plurality of fingers includes a first finger. The plurality of fingers further includes a second finger, wherein the first finger and the second finger define a first cavity for receiving the pipe. The plurality of fingers further includes a third finger, wherein the second

finger is between the first finger and the third finger, and the third finger and the second finger define a second cavity. The buffer further includes a roller on the base opposite the plurality of fingers. In some embodiments, a depth of the first cavity in a first direction is less than a depth of the second cavity in the first direction, and the first direction is parallel to a bottom surface of the third finger. In some embodiments, a material of the base has a coefficient of friction of less than 0.1. In some embodiments, the base includes polytetrafluoroethylene (PTFE), perfluoroalkoxy (PFA), fluorinated polyethylene propylene (FPEP), polyvinylidene fluoride (PVDF), polysulfone, or polyether ether ketone (PEEK). In some embodiments, the second cavity is configured to a cable different from the pipe. In some embodiments, the buffer further includes a ring surrounding the roller. In some embodiments, the roller has a hemispherical shape.

(47) An aspect of this description relates to a buffer for holding a pipe adapted to transport a fluid. The buffer includes a base. The buffer further includes a plurality of fingers extending outwardly from a first side of the base. The plurality of fingers includes a first finger, wherein the first finger includes a first protrusion extending from an end of the first finger distal from the base. The plurality of fingers further includes a second finger, wherein the second finger includes a second protrusion extending from a top surface, and the first finger and the second finger define a first cavity for receiving the pipe. The buffer further includes a roller on the base opposite the plurality of fingers. In some embodiments, the plurality of fingers further includes a third finger, wherein the second finger is between the first finger and the third finger, and the third finger and the second finger define a second cavity. In some embodiments, the first protrusion and the second protrusion define an outer edge of the first cavity. In some embodiments, a length of the second finger is greater than a length of the first finger.

(48) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A buffer for holding a pipe adapted to transport a fluid, comprising: a base; a plurality of fingers extending outwardly from a first side of the base, wherein each of the plurality of fingers is one-piece with the base, a first finger of the plurality of fingers and a second finger of the plurality of fingers define a cavity for receiving the pipe, the first finger extends outwardly from a first end of the base, and the second finger extends outwardly from a central portion of the base, and a third finger of the plurality of fingers extends outwardly from a second end of the base opposite the first end, wherein the second finger is between the first finger and the third finger; and at least one roller on a second side of the base opposite the first side.
2. The buffer of claim 1, wherein a length of the first finger is less than a length of at least one of the second finger or the third finger.
3. The buffer of claim 1, wherein the first finger comprises a first protrusion extending from an end of the first finger distal from the base, wherein the first protrusion extends toward the second finger.
4. The buffer of claim 3, wherein the second finger comprises a second protrusion extending from a top surface of the second finger, and the second protrusion extends toward the first finger.
5. The buffer of claim 4, wherein the second protrusion is aligned with the first protrusion.
6. The buffer of claim 1, wherein the second finger and the third finger define a second cavity

therebetween.

7. The buffer of claim 6, wherein the second cavity comprises a curved wall adjacent to the base.
 8. The buffer of claim 1, further comprising a ring surrounding the at least one roller.
 9. The buffer of claim 1, wherein the at least one roller includes a plurality of rollers.
 10. A buffer for holding a pipe adapted to transport a fluid, comprising: a base; a plurality of fingers extending outwardly from a first side of the base, wherein each of the plurality of fingers is one-piece with the base, and the plurality of fingers comprises: a first finger, a second finger, wherein the first finger and the second finger define a first cavity for receiving the pipe, and a third finger, wherein the second finger is between the first finger and the third finger, and the third finger and the second finger define a second cavity; and a roller on the base opposite the plurality of fingers.
 11. The buffer of claim 10, wherein a depth of the first cavity in a first direction is less than a depth of the second cavity in the first direction, and the first direction is parallel to a bottom surface of the third finger.
 12. The buffer of claim 10, wherein a material of the base has a coefficient of friction of less than 0.1.
 13. The buffer of claim 10, wherein the base comprises polytetrafluoroethylene (PTFE), perfluoroalkoxy (PFA), fluorinated polyethylene propylene (FPEP), polyvinylidene fluoride (PVDF), polysulfone, or polyether ether ketone (PEEK).
 14. The buffer of claim 10, wherein the second cavity is configured to a cable different from the pipe.
 15. The buffer of claim 10, further comprising a ring surrounding the roller.
 16. The buffer of claim 10, wherein the roller has a hemispherical shape.
 17. A buffer for holding a pipe adapted to transport a fluid, comprising: a base; a plurality of fingers extending outwardly from a first side of the base, wherein the plurality of fingers comprises: a first finger, wherein the first finger comprises a first protrusion extending from an end of the first finger distal from the base, a second finger, wherein the second finger comprises a second protrusion extending from a top surface, and the first finger and the second finger define a first cavity for receiving the pipe, and a roller on the base opposite the plurality of fingers.
 18. The buffer of claim 17, wherein the plurality of fingers further comprises a third finger, wherein the second finger is between the first finger and the third finger, and the third finger and the second finger define a second cavity.
 19. The buffer of claim 17, wherein the first protrusion and the second protrusion define an outer edge of the first cavity.
 20. The buffer of claim 17, wherein a length of the second finger is greater than a length of the first finger.
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